Chapter 2. ARMA Models

Autoregressive and moving average processes, PACF

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2.1 Introduction

Motivation. The autoregressive moving average (ARMA) models are the workhorse of stationary time series modelling. They arise from two natural ways of building a dependent stationary series out of i.i.d. noise: adding up neighbouring noise terms, and regressing the series on its own past.

Let $\epsilon_1, \epsilon_2, \dots$ be i.i.d. with mean 0 and variance σ^2.

2.1.1 Summing noise: MA(1)

For a real θ put

$$X_t = \epsilon_t + \theta\epsilon_{t-1}. \tag{2.1}$$

Then $EX_t = 0$, $\mathrm{Var}(X_t) = (1+\theta^2)\sigma^2$, and since $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$,

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1},\, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \geq 2, \end{cases}$$

which does not depend on t. Thus $\{X_t\}$ is stationary; it is the *moving average model of order* 1, MA(1), the simplest stationary model.

2.1.2 Regressing on the past: AR(1)

The simplest regression of the series on itself is

$$X_t = \phi X_{t-1} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2). \tag{2.2}$$

Does a stationary series satisfy (2.2)?

Case $\phi = 1$ (random walk). Iterating gives $X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t$, so $\mathrm{Var}(X_t - X_0) = t\sigma^2$. If $\{X_t\}$ were stationary with ACVF γ, then

$$\mathrm{Var}(X_t - X_0) = 2\gamma(0) - 2\gamma(t), \tag{2.3}$$

so $2\gamma(0) - 2\gamma(t) = t\sigma^2$. The left side is at most $4\gamma(0)$ for all t by Theorem 1.2.6(3), while the right side diverges, a contradiction. Hence no stationary solution exists; the process with $\phi = 1$ is the *random walk*, whose paths wander with no fixed level. The case $\phi = -1$ is similar: then $X_{2t} - X_0 = \sum_{j=1}^{2t} (-1)^j \epsilon_j$ has variance $2t\sigma^2$, again contradicting stationarity.

Case $|\phi| < 1$. Suppose a stationary solution exists. Iterating (2.2),

$$X_t = \phi X_{t-1} + \epsilon_t = \phi(\phi X_{t-2} + \epsilon_{t-1}) + \epsilon_t = \cdots = \sum_{j=0}^{n-1} \phi^j \epsilon_{t-j} + \phi^n X_{t-n}. \tag{2.4}$$

Since $E(\phi^n X_{t-n})^2 = \phi^{2n} EX_0^2 \to 0$, the only candidate is

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-i} \tag{2.5}$$

(the series converges in mean square and almost surely since $\sum |\phi|^i < \infty$). It is indeed a solution, because

$$\phi X_{t-1} + \epsilon_t = \sum_{i=0}^{\infty} \phi^{i+1} \epsilon_{t-1-i} + \epsilon_t = \sum_{j=1}^{\infty} \phi^j \epsilon_{t-j} + \epsilon_t = X_t,$$

and it is stationary: $EX_t = 0$ and

$$\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}\Big(\sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}, \sum_{k=0}^{\infty} \phi^k \epsilon_{t+h-k}\Big) = \sum_{j=0}^{\infty} \phi^j \phi^{j+h} \sigma^2 = \phi^h \frac{\sigma^2}{1-\phi^2}, \quad h \geq 0.$$

The model (2.2) is the *AR(1) model*. These two constructions are the building blocks of the general models below.

2.2 Autoregressive models

Motivation. An AR(p) model regresses X_t on p of its own lags. Whether it has a stationary solution is decided by where the roots of a polynomial lie, and the backshift operator turns the model into polynomial algebra.

Definition 2.2.1 (Autoregressive model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.6}$$

then $\{X_t\}$ follows an *autoregressive model of order* p, written $\{X_t\} \sim AR(p)$.

Define the *characteristic polynomial* $\phi : \mathbb{C} \to \mathbb{C}$,

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p. \tag{2.7}$$

If every root of $\phi(z) = 0$ satisfies $|z| > 1$, then $1/\phi(z)$ has a power series expansion valid on $|z| \leq 1$,

$$\frac{1}{\phi(z)} = \sum_{i=0}^{\infty} a_i z^i, \tag{2.8}$$

and the stationary solution of (2.6) is

$$X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}. \tag{2.9}$$

Thus X_t is a function of ϵ_s, $s \leq t$, only, so X_t is uncorrelated with (for i.i.d. noise, independent of) ϵ_s, $s > t$. Such a representation is called *causal*. These facts are proved in Proposition 2.2.B below.

2.2.1 The backshift operator

The *backshift operator* B acts by $BX_t = X_{t-1}$, and on functions by $Bf(t) = f(t-1)$; thus $B^j X_t = X_{t-j}$. For power series $a(z) = \sum_{i \geq 0} a_i z^i$, $b(z) = \sum_{j \geq 0} b_j z^j$ with $\sum |a_i| < \infty$, $\sum |b_j| < \infty$, put

$$c(z) = (ab)(z) = a(z)b(z) = \sum_{k=0}^{\infty} c_k z^k, \qquad c_k = \sum_{i=0}^{k} a_i b_{k-i}.$$

For a stationary $\{X_t\}$ one then has

$$a(B)\big(b(B)X_t\big) = (ab)(B)X_t = b(B)\big(a(B)X_t\big), \tag{2.10}$$

and all these series are stationary. With this notation (2.6) reads $\phi(B)X_t = \epsilon_t$, and (2.9) reads

$$X_t = \frac{1}{\phi(B)}\epsilon_t, \qquad \frac{1}{\phi(B)} = \sum_{i=0}^{\infty} a_i B^i. \tag{2.11}$$

Lemma 2.2.A (Operator calculus; supplement)

Let $\{X_t\}$ satisfy $\sup_t E|X_t| < \infty$ (e.g. stationary), and let $\sum |a_i| < \infty$, $\sum |b_j| < \infty$. Then $b(B)X_t = \sum_j b_j X_{t-j}$ converges absolutely a.s. and in L^1, (2.10) holds a.s., and if $\{X_t\}$ is stationary so is $\{b(B)X_t\}$, with ACVF $\sum_{j,k} b_j b_k \gamma_X(h+j-k)$.

Proof. [Supplement]

Step 1. Let $M = \sup_t E|X_t|$. By monotone convergence $E\sum_j |b_j||X_{t-j}| \le M \sum_j |b_j| < \infty$, so the series converges absolutely a.s. and in L^1. If $\{X_t\}$ is stationary, then $E|X_t| \le (EX_t^2)^{1/2}$ is bounded, and the same argument in L^2 (Minkowski: $\| \sum_j |b_j||X_{t-j}| \|_2 \le \sum_j |b_j| \, \|X_0\|_2$) gives mean-square convergence.

Step 2. Since $\sum_i \sum_j |a_i||b_j| E|X_{t-i-j}| \le M \sum_i |a_i| \sum_j |b_j| < \infty$, the double series $\sum_{i,j} a_i b_j X_{t-i-j}$ converges absolutely a.s., so it may be summed in any order (Fubini). Summing over j first gives $a(B)(b(B)X_t)$; grouping by $k = i + j$ gives $\sum_k c_k X_{t-k} = (ab)(B)X_t$. Since $ab = ba$, the third expression in (2.10) follows by symmetry.

Step 3. $E\, b(B)X_t = \mu \sum b_j$, and by L^2 convergence covariances can be taken termwise: $\mathrm{Cov}(b(B)X_t, b(B)X_{t+h}) = \sum_{j,k} b_j b_k \gamma_X(h+j-k)$, which is free of t and absolutely convergent. ■

Proposition 2.2.B (Causal stationary solution; supplement)

Suppose $\phi(z) \neq 0$ for all $|z| \le 1$. Then:
(1) there are $r > 1$ and $C < \infty$ with $1/\phi(z) = \sum_{i \ge 0} a_i z^i$ for $|z| < r$ and $|a_i| \le C r_0^{-i}$ for any fixed $1 < r_0 < r$;
(2) $X_t = \sum_i a_i \epsilon_{t-i}$ is a stationary solution of (2.6);
(3) it is the only stationary solution.

Proof. [Supplement]

Step 1. The roots of the polynomial ϕ are finitely many and all have modulus > 1; let $r > 1$ be the smallest modulus. Then $1/\phi$ is analytic on the disc $|z| < r$ and has a Taylor series $\sum a_i z^i$ there. For $1 < r_0 < r$, Cauchy's estimate on the circle $|z| = r_0$ gives $|a_i| \le r_0^{-i} \max_{|z|=r_0} |1/\phi(z)| = C r_0^{-i}$. In particular $\sum |a_i| < \infty$.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is well defined and stationary, and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \epsilon_t$ because $\phi(z)a(z) = 1$ for $|z| \le 1$, so the coefficient sequence of ϕa is $(1, 0, 0, \dots)$.

Step 3. Let $\{Y_t\}$ be any stationary solution, $\phi(B)Y_t = \epsilon_t$. Applying $a(B)$ and using the Lemma,

$$a(B)\epsilon_t = a(B)(\phi(B)Y_t) = (a\phi)(B)Y_t = Y_t,$$

so $Y_t = X_t$ a.s. ■

For $p = 1$, $\phi(z) = 1 - \phi z$ has root $1/\phi$, which lies outside the unit circle iff $|\phi| < 1$, and $a_i = \phi^i$, recovering (2.5).

Remark 2.2.2 (Nonzero mean)

A stationary AR(p) model with mean μ is written

$$X_t - \mu - \phi_1(X_{t-1} - \mu) - \phi_2(X_{t-2} - \mu) - \cdots - \phi_p(X_{t-p} - \mu) = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2).$$

Example 2.2.3 (Variance of a stationary AR(1))

Let $X_t = \phi X_{t-1} + \epsilon_t$ with $|\phi| < 1$. Since X_{t-1} depends only on ϵ_s, $s \le t-1$, $\mathrm{Cov}(X_{t-1}, \epsilon_t) = 0$, so

$$\gamma(0) = \mathrm{Var}(\phi X_{t-1} + \epsilon_t) = \phi^2 \gamma(0) + \sigma^2, \qquad \text{hence} \qquad \gamma(0) = \frac{\sigma^2}{1-\phi^2}.$$

2.3 Moving average models

Motivation. MA(q) models are finite linear filters of white noise. They are always stationary, and their autocorrelation vanishes beyond lag q, a signature that identifies them in data.

Definition 2.3.1 (Moving average model)

If $\{X_t\}$ satisfies

$$X_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.12}$$

then $\{X_t\}$ follows a *moving average model of order* q, written $\{X_t\} \sim MA(q)$.

With the backshift operator,

$$X_t = \theta(B)\epsilon_t, \qquad \theta(z) = \sum_{i=0}^{q} \theta_i z^i, \quad \theta_0 = 1. \tag{2.13}$$

For MA(1), $X_t = \epsilon_t + \theta \epsilon_{t-1}$ with $\{\epsilon_t\} \sim WN(0, \sigma^2)$, we found in §2.1

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1}, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \ge 2. \end{cases} \tag{2.14}$$

In general, using $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$, for $h \ge 0$

$$\gamma(h) = \sum_{i=0}^{q} \sum_{j=0}^{q} \theta_i \theta_j \,\mathrm{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_{i=0}^{q-h} \theta_i \theta_{i+h},$$

the sum being empty when $h > q$. Hence, by symmetry in h,

$$\gamma(h) = \begin{cases} 0, & |h| \ge q+1, \\ (\theta_{|h|} + \theta_1 \theta_{1+|h|} + \cdots + \theta_{q-|h|}\theta_q)\sigma^2, & |h| \le q. \end{cases} \tag{2.15}$$

Since γ depends only on h, an MA(q) is always stationary, with mean 0 and variance $(1 + \theta_1^2 + \cdots + \theta_q^2)\sigma^2$. Its ACF is zero beyond lag q; conversely, a sample ACF that vanishes from some lag on suggests an MA model.

2.4 Autoregressive moving average models

Motivation. Combining an AR and an MA part gives models that describe rich dependence with few parameters. Two root conditions matter: causality (roots of ϕ) for stationarity, and invertibility (roots of θ) for recovering the noise from the data.

Definition 2.4.1 (ARMA model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.16}$$

then $\{X_t\}$ follows an *autoregressive moving average model of orders p and q*, written $\{X_t\} \sim ARMA(p, q)$. With $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$, (2.16) reads

$$\phi(B) X_t = \theta(B) \epsilon_t. \tag{2.17}$$

If $\{X_t\}$ is stationary with mean $\mu \neq 0$, X_t is replaced by $X_t - \mu$ in (2.17).

ARMA models came out of the experience that pure AR fits often need a large order. The *principle of parsimony*, prefer the model with fewer parameters among models explaining the data equally well, underlies all model selection, and parsimonious ARMA models are known to fit real data well.

As for AR(p), if $\phi(z) \neq 0$ for all $|z| \leq 1$ then $\{X_t\}$ is stationary and

$$X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}, \qquad \sum_{i=0}^{\infty} a_i z^i = \frac{\theta(z)}{\phi(z)}, \tag{2.18}$$

that is, by (2.10),

$$X_t = \frac{\theta(B)}{\phi(B)} \epsilon_t = \Big(\frac{\theta}{\phi}\Big)(B) \epsilon_t. \tag{2.19}$$

The ACVF of (2.18) decays to 0 geometrically as h grows; for this reason ARMA processes are called *short-memory processes*.

Proposition 2.4.A (Causal ARMA and geometric decay; supplement)

If $\phi(z) \neq 0$ for $|z| \leq 1$, then (2.18) is the unique stationary solution of (2.17), with $|a_i| \leq C r_0^{-i}$ for some $C < \infty$, $r_0 > 1$, and

$$\gamma(h) = \sigma^2 \sum_{i=0}^{\infty} a_i a_{i+|h|}, \qquad |\gamma(h)| \leq C' r_0^{-|h|}.$$

Proof. [Supplement]

Step 1. θ/ϕ is analytic on $|z| < r$ ($r > 1$ the smallest modulus of a root of ϕ), so the Cauchy estimate gives $|a_i| \leq C r_0^{-i}$ exactly as in Proposition 2.2.B.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is stationary and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \theta(B)\epsilon_t$, because $\phi a = \theta$ as power series. If Y is another stationary solution, then $Y_t = (\phi^{-1}\phi)(B) Y_t = \phi^{-1}(B)\theta(B)\epsilon_t = a(B)\epsilon_t$, where $\phi^{-1}(z) = 1/\phi(z)$ has absolutely summable coefficients.

Step 3. For $h \geq 0$, $\gamma(h) = \sum_{i,j} a_i a_j \operatorname{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_i a_i a_{i+h}$ (take $j = i + h$), and

$$|\gamma(h)| \leq \sigma^2 C^2 \sum_{i \geq 0} r_0^{-i} r_0^{-i-h} = \frac{\sigma^2 C^2}{1 - r_0^{-2}} r_0^{-h}.$$

■

2.4.1 Invertibility

If $\{X_t\}$ is a stationary ARMA process and $\theta(z) \neq 0$ for all $|z| \leq 1$, then

$$\sum_{j=0}^{\infty} \pi_j X_{t-j} = \epsilon_t, \qquad \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad \pi_0 = 1. \tag{2.20}$$

In this case $\{X_t\}$ is called *invertible*: the current noise, and hence the present value, is a linear combination of infinitely many past observations.

Proposition 2.4.B (Invertibility; supplement)

Let $\{X_t\}$ be a stationary solution of $\phi(B)X_t = \theta(B)\epsilon_t$ and suppose $\theta(z) \neq 0$ for $|z| \leq 1$. Then (2.20) holds a.s., with $|\pi_j|$ decaying geometrically.

Proof. [Supplement] As in Proposition 2.2.B, $1/\theta(z) = \sum_j \eta_j z^j$ with geometrically decaying η_j, and $\pi(z) = \phi(z)/\theta(z) = \phi(z)\eta(z)$ also has geometrically decaying coefficients (ϕ is a polynomial). By the Lemma applied to the stationary series $\{X_t\}$ and $\{\epsilon_t\}$,

$$\pi(B)X_t = \eta(B)(\phi(B)X_t) = \eta(B)(\theta(B)\epsilon_t) = (\eta\theta)(B)\epsilon_t = \epsilon_t.$$

■

Example 2.4.2 (One ACF, two MA(1) models)

The models

$$X_t = \epsilon_t + 2\epsilon_{t-1} \quad \text{(not invertible)}, \qquad X_t = \epsilon_t + \tfrac{1}{2}\epsilon_{t-1} \quad \text{(invertible)} \tag{2.21}$$

have the same ACF: $\rho(1) = \theta/(1+\theta^2) = 2/5$ for $\theta = 2$ and for $\theta = 1/2$, and $\rho(h) = 0$ for $|h| \geq 2$. (Generally θ and $1/\theta$ give the same MA(1) ACF; the noise variances differ so that the ACVFs also agree: σ^2 for $\theta = 2$ corresponds to $4\sigma^2$ for $\theta = 1/2$.)

Thus a single ACF corresponds to both an invertible and a non-invertible model, and the invertible one is preferred. The reason is prediction: given $X_1, \ldots, X_n$ with n large, (2.20) gives the easily computed predictor

$$\hat{X}_{n+1} = -\pi_1 X_n - \pi_2 X_{n-1} - \cdots - \pi_n X_1,$$

whereas no such representation is available in the non-invertible case.

The orders p, q must be determined from data. Too many parameters violate parsimony, too few cause *model bias*; criteria such as AIC and BIC are used (Chapter 4). The choice matters for inference: e.g. for an AR(p) with i.i.d. noise and $n > p$,

$$E(X_{n+1} \mid X_1, \ldots, X_n) = E(\phi_1 X_n + \cdots + \phi_p X_{n-p+1} + \epsilon_{n+1} \mid X_1, \ldots, X_n) = \phi_1 X_n + \cdots + \phi_p X_{n-p+1},$$

since ϵ_{n+1} is independent of $X_1, \ldots, X_n$ by causality; a wrong p yields a wrong predictor.

Example 2.4.3 (ACVF of a causal ARMA(1,1))

Let $X_t = \phi X_{t-1} + \epsilon_t + \theta\epsilon_{t-1}$, $|\phi| < 1$. Since X_{t-h} is uncorrelated with $\epsilon_t, \epsilon_{t-1}$ for $h \geq 2$,

$$\gamma(h) = \mathrm{Cov}(\phi X_{t-1} + \epsilon_t + \theta\epsilon_{t-1}, X_{t-h}) = \phi\gamma(h-1), \quad \text{so} \quad \gamma(h) = \phi^{h-1}\gamma(1), \ h \geq 1.$$

Using $\mathrm{Cov}(\epsilon_{t-1}, X_{t-1}) = \sigma^2$ and $\mathrm{Cov}(\epsilon_t, X_t) = \sigma^2$, $\mathrm{Cov}(\epsilon_{t-1}, X_t) = (\phi+\theta)\sigma^2$:

$$\begin{aligned} \gamma(1) &= \phi\gamma(0) + \theta\sigma^2, \\ \gamma(0) &= \phi^2\gamma(0) + \sigma^2 + \theta^2\sigma^2 + 2\phi\theta\sigma^2. \end{aligned}$$

Hence

$$\gamma(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1-\phi^2}, \qquad \gamma(1) = \frac{\sigma^2(\phi+\theta)(1+\phi\theta)}{1-\phi^2}.$$

2.5 The partial autocorrelation function

Motivation. The ACF identifies MA orders but not AR orders. The partial autocorrelation removes the linear effect of the intermediate observations and cuts off after lag p for an AR(p), giving the complementary identification tool.

Let $\{X_t\}$ be stationary with mean 0. To measure the correlation of X_1 and X_{1+k} after removing the effect of $X_2, \dots, X_k$, let

$$F = \Big\{ \sum_{i=2}^{k} a_i X_i : a_i \in \mathbb{R} \Big\},$$

and let $P_F Y$ denote the *projection* of Y onto F: the element of F with $E\big[(Y - P_F Y) X_i\big] = 0$ for all $i = 2, \dots, k$. ($P_F Y$ is the best linear predictor of Y from $X_2, \dots, X_k$; for $k = 1$, $F = \{0\}$.)

Definition 2.5.1 (Partial autocorrelation function)

For $k \geq 1$,

$$\phi_{kk} = \mathrm{Corr}(X_{k+1} - P_F X_{k+1},\ X_1 - P_F X_1)$$

is the *partial autocorrelation function* (PACF); for fixed k, ϕ_{kk} is the partial autocorrelation coefficient at lag k.

2.5.1 Computing the PACF

Let ρ be the ACF. Solve

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(h-1) \\ \rho(1) & \rho(0) & \cdots & \rho(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(h-1) & \rho(h-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_{h1} \\ \phi_{h2} \\ \vdots \\ \phi_{hh} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(h) \end{pmatrix}. \tag{2.22}$$

The last component ϕ_{hh} of the solution is the PACF at lag h. By Cramer's rule, writing $R_h = [\rho(i-j)]_{i,j=1}^{h}$ and R_h^* for R_h with its last column replaced by $(\rho(1), \dots, \rho(h))^T$,

$$\phi_{hh} = \frac{\det R_h^*}{\det R_h}.$$

Proposition 2.5.A (PACF via the prediction equations; supplement)

Let $\{X_t\}$ be stationary, mean 0, with $\Gamma_h = [\gamma(i-j)]_{i,j=1}^{h}$ nonsingular. Then the best linear predictor of X_{h+1} from $X_1, \dots, X_h$ is $\sum_{j=1}^{h} \phi_{hj} X_{h+1-j}$ with coefficients solving (2.22), and its coefficient ϕ_{hh} on X_1 equals the PACF of Definition 2.5.1.

Proof. [Supplement]

Step 1. A linear combination $\hat{X} = \sum_j c_j X_{h+1-j}$ is the projection of X_{h+1} onto $\mathrm{span}(X_1, \dots, X_h)$ iff $E[(X_{h+1} - \hat{X}) X_{h+1-i}] = 0$ for $i = 1, \dots, h$, i.e. $\sum_j c_j \gamma(i-j) = \gamma(i)$. Dividing by $\gamma(0)$ gives (2.22), uniquely solvable since Γ_h is nonsingular.

Step 2. Let $F = \mathrm{span}(X_2, \dots, X_h)$, $U = X_1 - P_F X_1 \perp F$ (here $\perp$ means uncorrelated). Then $\mathrm{span}(X_1, \dots, X_h) = F \oplus \mathrm{span}(U)$ orthogonally, so the projection of X_{h+1} is

$$P_F X_{h+1} + \frac{E(X_{h+1} U)}{E U^2}\, U = P_F X_{h+1} + \frac{E(X_{h+1} U)}{E U^2}\, (X_1 - P_F X_1).$$

Since $P_F X_{h+1}, P_F X_1 \in F$ do not involve X_1, the coefficient of X_1 is $\phi_{hh} = E(X_{h+1}U)/EU^2$ (coefficients are unique as Γ_h is nonsingular).

Step 3. Let $V = X_{h+1} - P_F X_{h+1}$. As $U \perp F$, $E(X_{h+1}U) = E(VU)$, so $\phi_{hh} = E(VU)/EU^2$. By stationarity the covariance matrix of $(X_2, \dots, X_h, X_{h+1})$ equals that of $(X_h, \dots, X_2, X_1)$ (both are $[\gamma(i-j)]$, as γ is even), so predicting X_{h+1} from $X_2, \dots, X_h$ and X_1 from $X_h, \dots, X_2$ have the same error variance: $EV^2 = EU^2$. Hence $\phi_{hh} = E(VU)/\sqrt{EV^2 EU^2} = \mathrm{Corr}(V, U)$. ■

Example 2.5.2 (PACF of an AR(1))

For a stationary AR(1), $\rho(h) = \phi^h$. At $h = 1$: $\phi_{11} = \rho(1) = \phi$. At $h = 2$:

$$\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix} \implies \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} = \frac{1}{1-\phi^2} \begin{pmatrix} 1 & -\phi \\ -\phi & 1 \end{pmatrix} \begin{pmatrix} \phi \\ \phi^2 \end{pmatrix} = \begin{pmatrix} \phi \\ 0 \end{pmatrix},$$

so $\phi_{22} = 0$; likewise $\phi_{hh} = 0$ for all $h \geq 2$.

Proposition 2.5.B (PACF of a causal AR(p); supplement)

If $\{X_t\}$ is a causal AR(p) with $\phi_p \neq 0$ and $\sigma^2 > 0$, then $\phi_{pp} = \phi_p$ and $\phi_{hh} = 0$ for all $h > p$.

Proof. [Supplement]

Step 1. Γ_h is nonsingular for every h. Let $a \neq 0$ and let i be the largest index with $a_i \neq 0$. By causality $X_i = \epsilon_i + \sum_{k \geq 1} a'_k \epsilon_{i-k}$ (the leading coefficient of $1/\phi$ is 1), and $X_1, \dots, X_{i-1}$ are functions of ϵ_u, $u < i$. Hence $Y = \sum_{j \leq i} a_j X_j = a_i \epsilon_i + W$ with W uncorrelated with ϵ_i, so $a^T \Gamma_h a = \mathrm{Var}(Y) \geq a_i^2 \sigma^2 > 0$.

Step 2. Let $h \geq p$. The candidate $\hat{X} = \phi_1 X_h + \cdots + \phi_p X_{h+1-p} \in \mathrm{span}(X_1, \dots, X_h)$ has error $X_{h+1} - \hat{X} = \epsilon_{h+1}$, which is uncorrelated with $X_1, \dots, X_h$ by causality (each X_s is a function of ϵ_u, $u \leq s$). Hence $\hat{X}$ is the projection, and by uniqueness its coefficients are $(\phi_1, \dots, \phi_p, 0, \dots, 0)$.

Step 3. By Proposition 2.5.A, ϕ_{hh} is the coefficient on $X_1 = X_{h+1-h}$: it equals ϕ_p when $h = p$ and 0 when $h > p$. ■

Thus the PACF of an AR(p) vanishes after lag p, while the ACF of an MA(q) vanishes after lag q; the opposite functions decay geometrically. Figure 1 illustrates this duality, which is the basis for choosing p and q from data (Chapters 3 and 4).

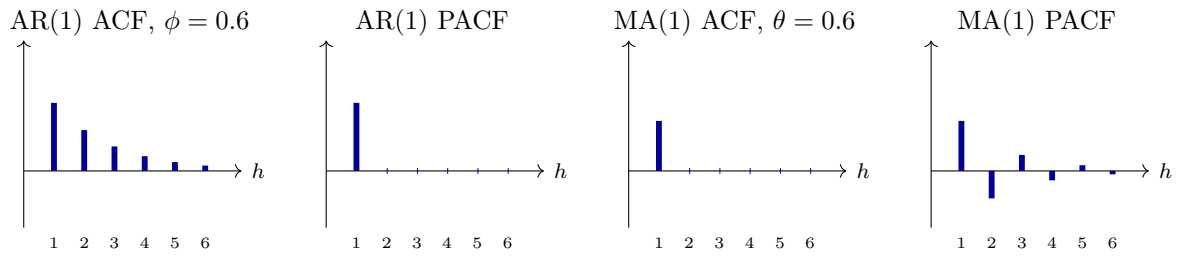


Figure 1: ACF and PACF of an AR(1) and an MA(1). The AR(1) PACF and the MA(1) ACF cut off after lag 1; the other two decay geometrically.

2.6 Selected exercises

Motivation. These textbook exercises go beyond the propositions above: sums of MA processes, models that share an ACF, and an exact invertibility criterion for ARMA(1,1). Covariance formulas

already proved in the text (the ACVF of a linear process, the operator calculus, the MA(q) ACVF) are not repeated.

Exercise 2.8 (Sum of independent MA(1) and MA(2))

Let $X_t = \epsilon_t + a\epsilon_{t-1}$ with $\{\epsilon_t\} \sim IID(0, \sigma_1^2)$ and $Y_t = \eta_t + b_1\eta_{t-1} + b_2\eta_{t-2}$ with $\{\eta_t\} \sim IID(0, \sigma_2^2)$, the two noise sequences being independent, and $\sigma_1^2 + \sigma_2^2 > 0$. Show that $Z_t = X_t + Y_t$ can be regarded as an MA(2) model: there are a white noise $\{u_t\} \sim WN(0, v)$ with $v > 0$ and constants c_1, c_2 such that

$$Z_t = u_t + c_1 u_{t-1} + c_2 u_{t-2} \qquad \text{for all } t.$$

Solution. All random variables below lie in L^2 with the inner product $E(UV)$; "$\perp$" means uncorrelated. Let H_t be the closed linear span of $\{Z_s : s \le t\}$ and P_t the orthogonal projection onto H_t.

Step 1. (Z is stationary with ACVF vanishing beyond lag 2.) By (2.15), $\gamma_X(h) = 0$ for $|h| \ge 2$ and $\gamma_Y(h) = 0$ for $|h| \ge 3$. Since $\mathrm{Cov}(X_s, Y_u) = 0$ for all s, u, Exercise 1.7 gives that Z is stationary with mean 0 and $\gamma_Z = \gamma_X + \gamma_Y$; hence $\gamma_Z(h) = 0$ for $|h| \ge 3$ and $\gamma_Z(0) = (1+a^2)\sigma_1^2 + (1+b_1^2+b_2^2)\sigma_2^2 > 0$. In particular $Z_t \perp Z_s$ for $s \le t-3$, so $Z_t \perp H_{t-3}$.

Step 2. (Innovations.) Put $u_t = Z_t - P_{t-1}Z_t$. Then $u_t \in H_t$ and $u_t \perp H_{t-1}$. For $s < t$, $u_s \in H_s \subseteq H_{t-1}$, so $E(u_s u_t) = 0$.

Step 3. (Shift invariance.) For $m \ge 1$ let $\alpha_m = (\alpha_{m,1}, \dots, \alpha_{m,m})^T$ be the minimum-norm solution of $\Gamma_m \alpha_m = (\gamma_Z(1), \dots, \gamma_Z(m))^T$, $\Gamma_m = [\gamma_Z(i-j)]_{i,j=1}^m$. As in Step 1 of the proof of Proposition 2.5.A, $\sum_i \alpha_{m,i} Z_{t-i}$ is the projection of Z_t onto $\mathrm{span}(Z_{t-1}, \dots, Z_{t-m})$, and α_m depends on γ_Z only, not on t. These subspaces increase and their union is dense in H_{t-1}, so the projections converge in L^2 to $P_{t-1}Z_t$. By continuity of the inner product, every covariance $E(u_t u_s)$, $E(u_t Z_s)$ is the limit of a covariance of finite linear combinations of the Z's with t-free coefficients, hence depends on $t - s$ only. In particular $v = Eu_t^2$ does not depend on t, and $\{u_t\} \sim WN(0, v)$ by Step 2.

Step 4. ($v > 0$.) If $v = 0$, then $Z_s = P_{s-1}Z_s \in H_{s-1}$ for every s, so $H_s = H_{s-1}$ for all s and $H_t = H_{t-3}$. By Step 1, $Z_t \perp H_{t-3} = H_t \ni Z_t$, so $EZ_t^2 = 0$, contradicting $\gamma_Z(0) > 0$.

Step 5. (Orthogonal decomposition.) Since $Z_s = u_s + P_{s-1}Z_s$ with $P_{s-1}Z_s \in H_{s-1}$, we have $H_s = H_{s-1} \oplus \mathrm{span}(u_s)$ (the sum of a closed subspace and an orthogonal one-dimensional subspace is closed). Applying this for $s = t-1$ and $s = t-2$,

$$H_{t-1} = H_{t-3} \oplus \mathrm{span}(u_{t-2}) \oplus \mathrm{span}(u_{t-1}).$$

Step 6. (Conclusion.) Projecting Z_t onto the three orthogonal pieces,

$$Z_t = u_t + P_{t-1}Z_t = u_t + \frac{E(Z_t u_{t-1})}{v}\, u_{t-1} + \frac{E(Z_t u_{t-2})}{v}\, u_{t-2} + P_{H_{t-3}} Z_t.$$

The last term is 0 because $Z_t \perp H_{t-3}$, and by Step 3 the coefficients $c_j = E(Z_t u_{t-j})/v$ do not depend on t. This is the claimed MA(2) representation. Moreover $u_{t-2} - Z_{t-2} \in H_{t-3} \perp Z_t$ gives $c_2 = \gamma_Z(2)/v = b_2\sigma_2^2/v$, so the order is exactly 2 when $b_2\sigma_2^2 \neq 0$.

The driving noise $\{u_t\}$ is white noise, in general not i.i.d.; "MA(2)" is meant at the level of second-order properties. ■

Exercise 2.9 (Two filters with the same ACF)

Let $\{Y_t\}$ be stationary with mean 0 and ACVF γ_Y, not a.s. zero, and let

$$X_t = Y_t - 0.4Y_{t-1}, \qquad W_t = Y_t - 2.5Y_{t-1}.$$

(a) Express the ACVFs of $\{X_t\}$ and $\{W_t\}$ through γ_Y.
(b) Show that $\{X_t\}$ and $\{W_t\}$ have the same ACF.

Solution.

(a) For $c \in \mathbb{R}$ and $V_t = Y_t - cY_{t-1}$, bilinearity and stationarity of Y give

$$\begin{aligned}\mathrm{Cov}(V_t, V_{t+h}) &= \gamma_Y(h) - c\,\gamma_Y(h-1) - c\,\gamma_Y(h+1) + c^2\gamma_Y(h) \\ &= (1+c^2)\gamma_Y(h) - c\{\gamma_Y(h-1) + \gamma_Y(h+1)\},\end{aligned}$$

free of t. With $c = 0.4$ and $c = 2.5$,

$$\begin{aligned}\gamma_X(h) &= 1.16\,\gamma_Y(h) - 0.4\{\gamma_Y(h-1) + \gamma_Y(h+1)\}, \\ \gamma_W(h) &= 7.25\,\gamma_Y(h) - 2.5\{\gamma_Y(h-1) + \gamma_Y(h+1)\}.\end{aligned}$$

(b)

Step 1. Since $2.5 = 1/0.4$, we have $7.25 = 6.25 \times 1.16$ and $2.5 = 6.25 \times 0.4$, so $\gamma_W(h) = 6.25\,\gamma_X(h)$ for all h. (In general $(1 + c^{-2}) = c^{-2}(1 + c^2)$ and $c^{-1} = c^{-2}c$.)

Step 2. The ACF is defined if $\gamma_X(0) > 0$. Suppose $\gamma_X(0) = 0$, i.e. $Y_t = 0.4Y_{t-1}$ a.s. for all t. Iterating, $Y_t = 0.4^n Y_{t-n}$, so $\gamma_Y(0) = 0.16^n\gamma_Y(0)$ for every n, forcing $\gamma_Y(0) = 0$, i.e. $Y_t = 0$ a.s., which is excluded.

Step 3. Hence $\rho_W(h) = \gamma_W(h)/\gamma_W(0) = 6.25\gamma_X(h)/(6.25\gamma_X(0)) = \rho_X(h)$.

This generalizes Example 2.4.2: the filters $1 - cB$ and $1 - c^{-1}B$ give the same ACF. ■

Exercise 2.10 (Invertibility of ARMA(1,1))

Let $\{X_t\}$ be the stationary solution of $X_t - \phi X_{t-1} = \epsilon_t + \theta\epsilon_{t-1}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, $\sigma^2 > 0$, $|\phi| < 1$. Find the condition for $\{X_t\}$ to be invertible, i.e. for $\epsilon_t = \sum_{j\geq 0} \pi_j X_{t-j}$ a.s. for some $\sum_j |\pi_j| < \infty$.

Solution. The condition is $|\theta| < 1$.

(a) Sufficiency.

Step 1. If $|\theta| < 1$, the root $-1/\theta$ of $\theta(z) = 1 + \theta z$ lies outside the unit disc, so Proposition 2.4.B applies and $\{X_t\}$ is invertible.

Step 2. Explicitly, for $|z| < 1/|\theta|$,

$$\frac{\phi(z)}{\theta(z)} = (1 - \phi z)\sum_{k\geq 0}(-\theta)^k z^k = 1 + \sum_{j\geq 1}\{(-\theta)^j - \phi(-\theta)^{j-1}\}z^j,$$

so $\pi_0 = 1$ and $\pi_j = -(\phi + \theta)(-\theta)^{j-1}$ for $j \geq 1$, which is absolutely summable.

(b) Necessity. Suppose $\epsilon_t = \pi(B)X_t$ with $\sum|\pi_j| < \infty$ and $|\theta| \geq 1$.

Step 1. By Proposition 2.4.A, $X_t = a(B)\epsilon_t$ with $\sum|a_i| < \infty$ and $a(z) = \theta(z)/\phi(z)$ for $|z| \leq 1$. By Lemma 2.2.A, $\epsilon_t = \pi(B)a(B)\epsilon_t = c(B)\epsilon_t$ with $c = \pi a$ (Cauchy product), $\sum|c_k| < \infty$.

Step 2. Taking the covariance of $\epsilon_t = \sum_k c_k\epsilon_{t-k}$ with ϵ_{t-k} (termwise, by L^2 convergence) gives $\sigma^2\mathbb{1}\{k = 0\} = c_k\sigma^2$, so $c_0 = 1$ and $c_k = 0$ for $k \geq 1$. Since both series converge absolutely on $|z| \leq 1$, $\pi(z)a(z) = c(z) = 1$ there, i.e.

$$\pi(z)\,\theta(z) = \phi(z), \qquad |z| \leq 1.$$

Step 3. The point $z_0 = -1/\theta$ satisfies $|z_0| \leq 1$ and $\theta(z_0) = 0$, so $\phi(z_0) = 1 + \phi/\theta = 0$, i.e. $\phi = -\theta$. But then $|\phi| = |\theta| \geq 1$, contradicting $|\phi| < 1$. Hence $|\theta| < 1$ is necessary.

If $\phi = -\theta$ (common factor), then $|\theta| = |\phi| < 1$ and $a(z) = 1$, so $X_t = \epsilon_t$ is trivially invertible, consistent with the criterion. ■